

C.3 Wave phenomena

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **26 marks**

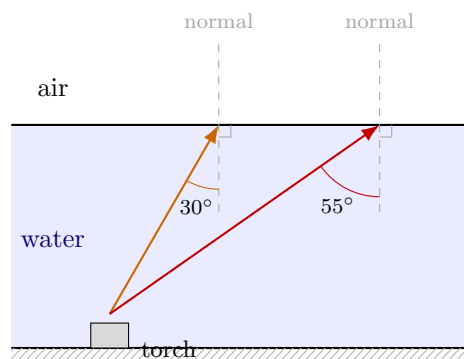
Refractive index of water = 1.33; of air = 1.00. Show your working.

Question 1 — The midnight pool torch

(11 marks)

During a night swim, a waterproof torch rests on the bottom of a pool, its beam angled up towards the surface.

- (a) State Snell's law, defining every symbol you use. [2]
- (b) The beam meets the surface at 30° to the normal. Calculate the angle at which it leaves the water. [3]
- (c) Show that the critical angle for the water–air surface is about 49° . [3]
- (d) The torch is tilted so the beam meets the surface at 55° . Describe and name what happens to the beam, and state one technology that depends on this effect. [3]



Question 2 — Laser night at the school hall

(7 marks)

For a science evening, red laser light of wavelength 650 nm passes through two narrow slits separated by 0.25 mm. Bright and dark fringes appear on a wall 4.0 m away.

- (a) State the condition on *path difference* for a bright fringe to form, and the condition for a dark fringe. [2]
- (b) Calculate the separation of neighbouring bright fringes on the wall. [3]
- (c) The red laser is swapped for a green one. State and explain what happens to the fringe separation. [2]

Question 3 — [HL] One slit is drama enough

(4 marks)

The same red light (650 nm) now passes through a single slit of width 0.10 mm.

- (a) Calculate the angle of the first diffraction minimum. [2]
- (b) Describe two ways the single-slit pattern differs from the double-slit fringe pattern. [2]

Question 4 — The pool hears you

(4 marks)

A lifeguard's whistle (2.0 kHz) is heard both in the air and by a diver underwater. Sound travels at 340 m s^{-1} in air and 1500 m s^{-1} in water.

- (a) State which property of the sound is unchanged as it crosses into the water, and explain why.
[2]

- (b) Calculate the ratio $\frac{\lambda_{\text{water}}}{\lambda_{\text{air}}}$. [2]

Answers

Question 1 — The midnight pool torch

- (a) $n_1 \sin \theta_1 = n_2 \sin \theta_2$, where n_1 and n_2 are the refractive indices of the two media and θ_1 , θ_2 are the angles between the ray and the *normal* in each medium.
- (b) $1.33 \sin 30^\circ = 1.00 \sin \theta \Rightarrow \sin \theta = 0.665 \Rightarrow \theta = 42^\circ$ (more precisely 41.7°). The ray bends *away* from the normal on entering the less dense medium.
- (c) At the critical angle the refracted ray grazes along the surface, $\theta_2 = 90^\circ$, so $\sin \theta_c = n_2/n_1 = 1/1.33 = 0.752$, giving $\theta_c = 48.8^\circ \approx 49^\circ$.
- (d) Since $55^\circ > \theta_c$, no light escapes: the beam undergoes *total internal reflection*, reflecting back down into the water with the angle of reflection equal to the 55° angle of incidence. Applications include optical fibres (telecommunications, endoscopy) and prismatic binoculars.

Question 2 — Laser night at the school hall

- (a) Bright fringe: path difference $= n\lambda$ for integer n . Dark fringe: path difference $= (n + \frac{1}{2})\lambda$.
- (b) $s = \frac{\lambda D}{d} = \frac{650 \times 10^{-9} \times 4.0}{0.25 \times 10^{-3}} = 1.0 \times 10^{-2} \text{ m} = 1.0 \text{ cm}$.
- (c) Green light has a shorter wavelength than red, and $s \propto \lambda$, so the fringes move *closer together*.

Question 3 — [HL] One slit is drama enough

- (a) $\theta = \frac{\lambda}{b} = \frac{650 \times 10^{-9}}{0.10 \times 10^{-3}} = 6.5 \times 10^{-3} \text{ rad}$, about 0.37° .
- (b) Two of: the central maximum is twice the width of the ones beside it; the intensity falls away sharply with order, so the side maxima are much dimmer; the pattern is therefore not a set of evenly spaced fringes of equal brightness, as the double slit gives.

Question 4 — The pool hears you

- (a) The *frequency* is unchanged. It is fixed by the source, and every wavefront arriving at the boundary must drive exactly one wavefront onwards into the water, so the rate cannot change. (The speed and wavelength both change.)
- (b) Since $\lambda = v/f$ with f common to both media, $\frac{\lambda_{\text{water}}}{\lambda_{\text{air}}} = \frac{v_{\text{water}}}{v_{\text{air}}} = \frac{1500}{340} = 4.4$.